

10) Draw a use case diagram for Library Management System is an automation system used to manage a library and the different resource management required in it like cataloguing of books, allowing check out and return of books, invoicing, user management, etc. The user can search for books details using few book properties (Book ID, Title, Author, and Publisher). Searching should return details about all the book copies that match the search query.

AIM :- To design a **Use Case Diagram** for a **Library Management System** that automates book cataloging, checkouts, returns, invoicing, and user management.

PROCEDURE :-

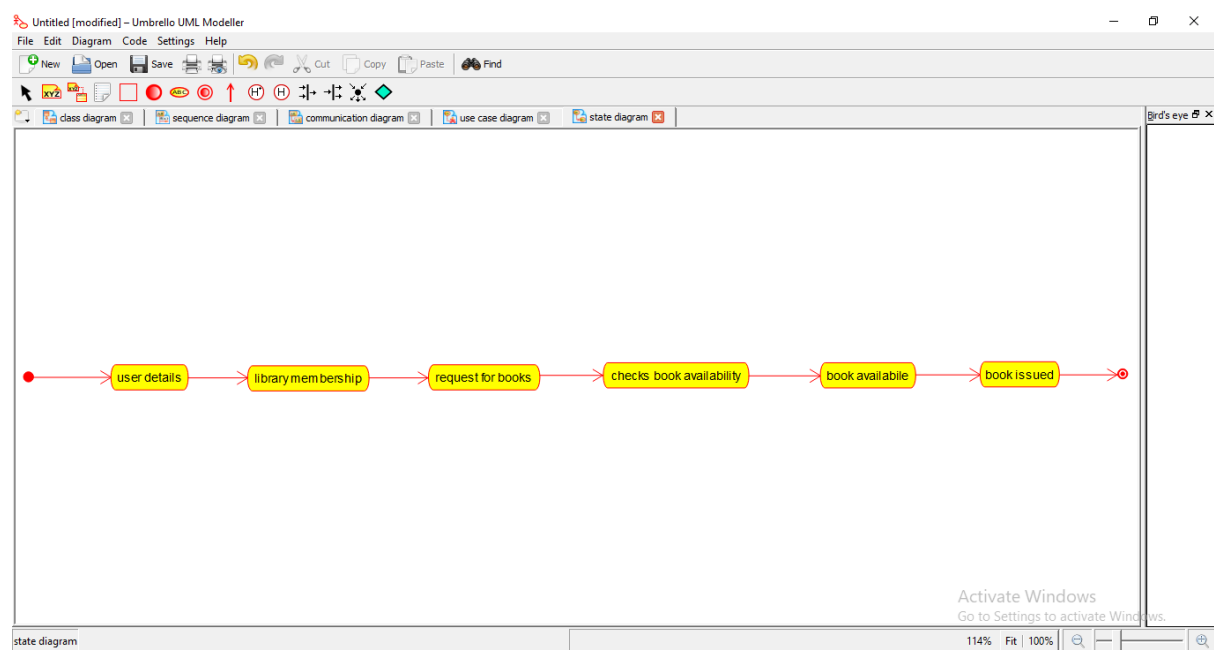
Identify the Key Actors.

Define the Major Use Cases (Functionalities).

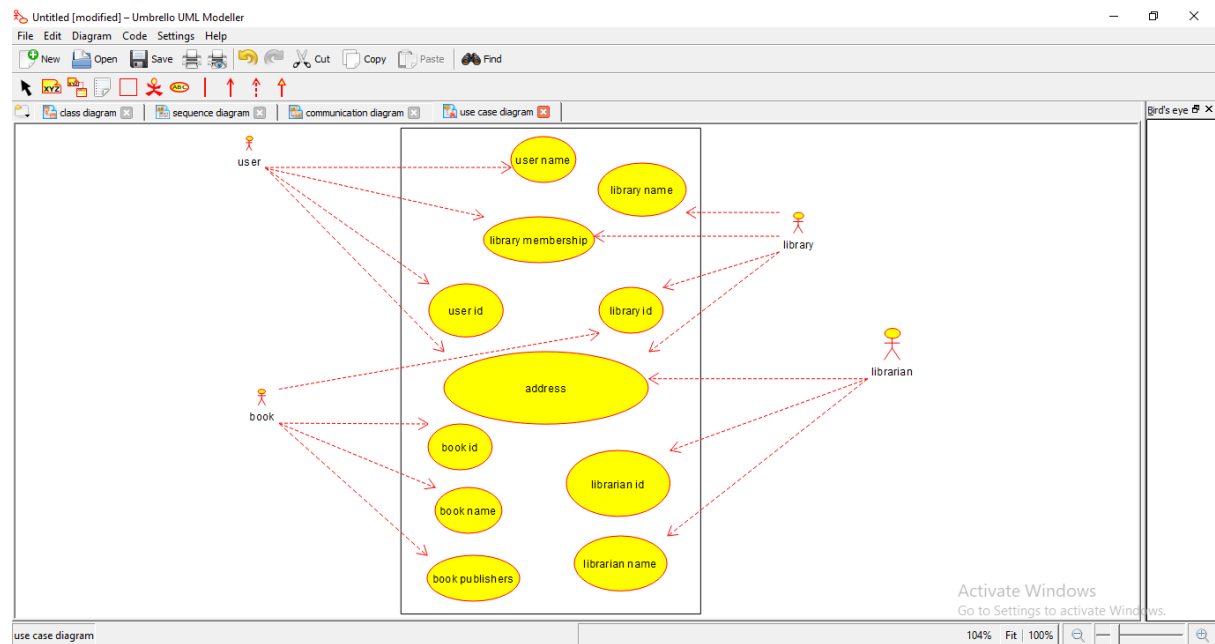
Draw the Use Case Diagram.

Label and Verify the Diagram.

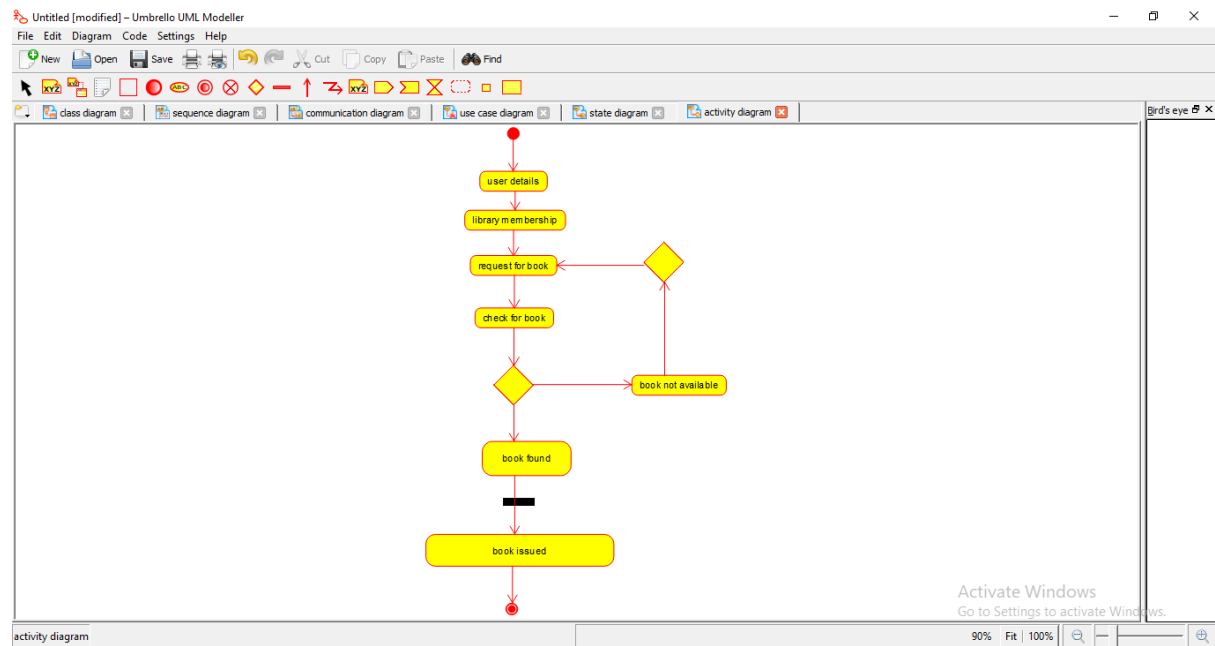
STATE DIAGRAM



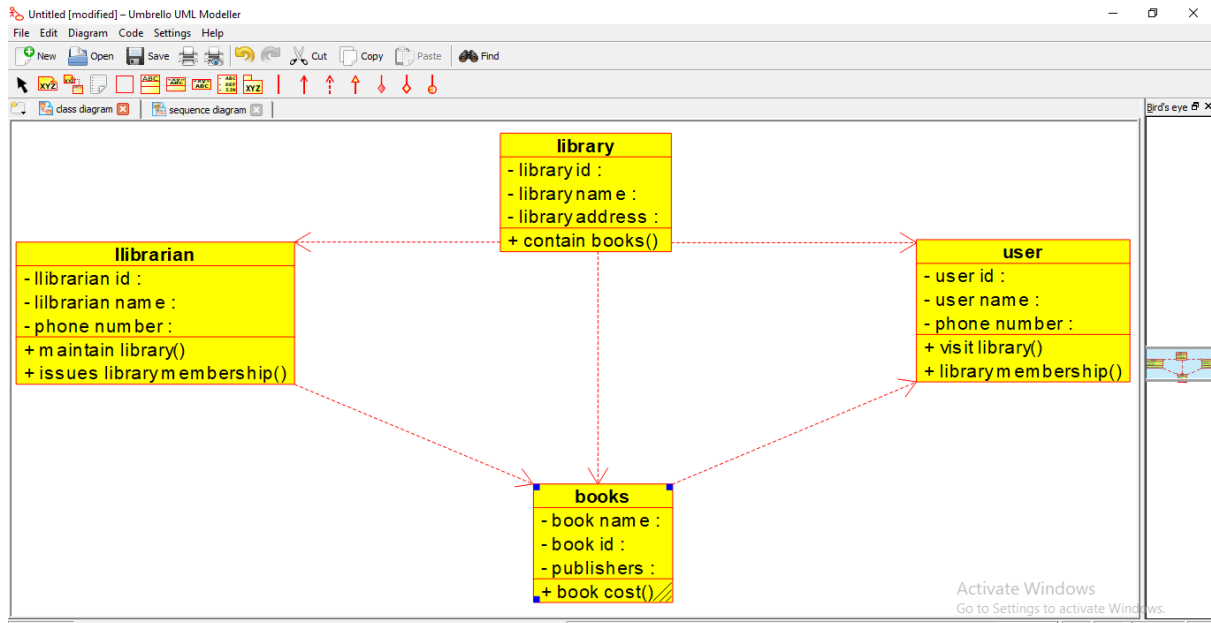
USE CASE DIAGRAM



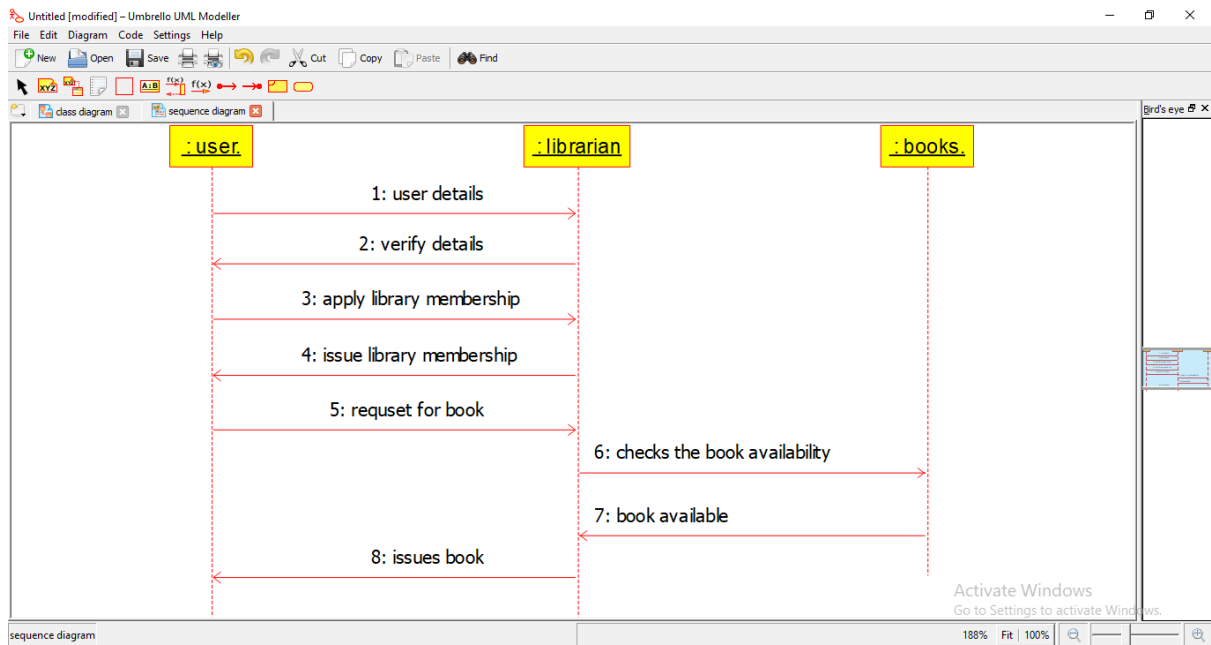
ACTIVITY DIAGRAM



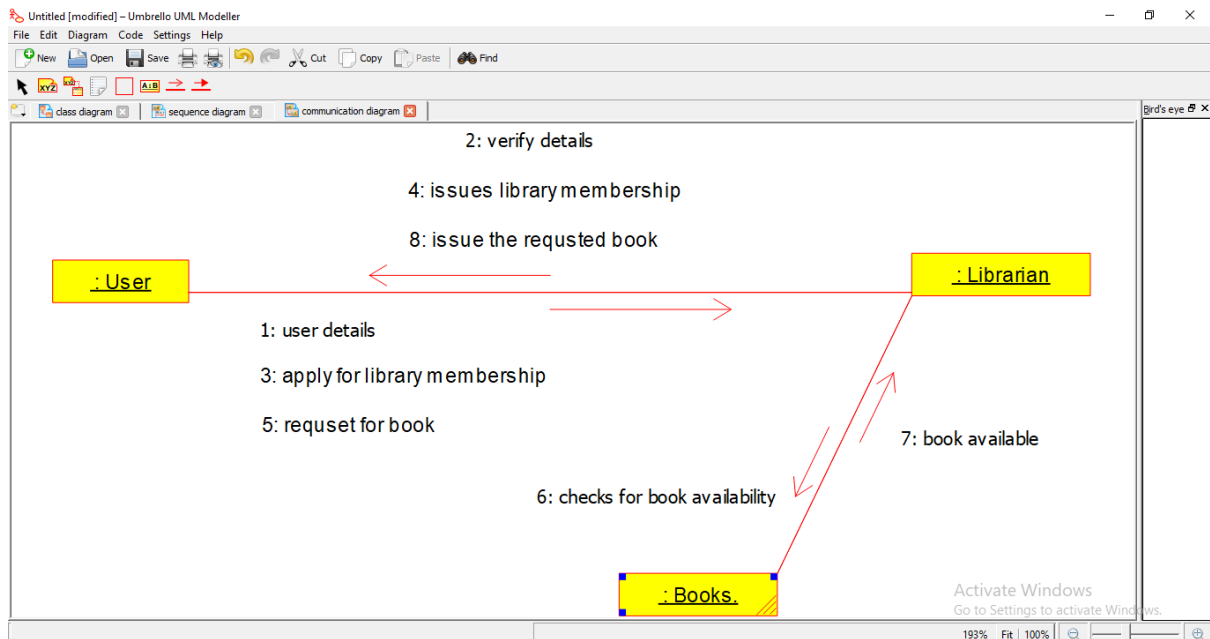
CLASS DIAGRAM



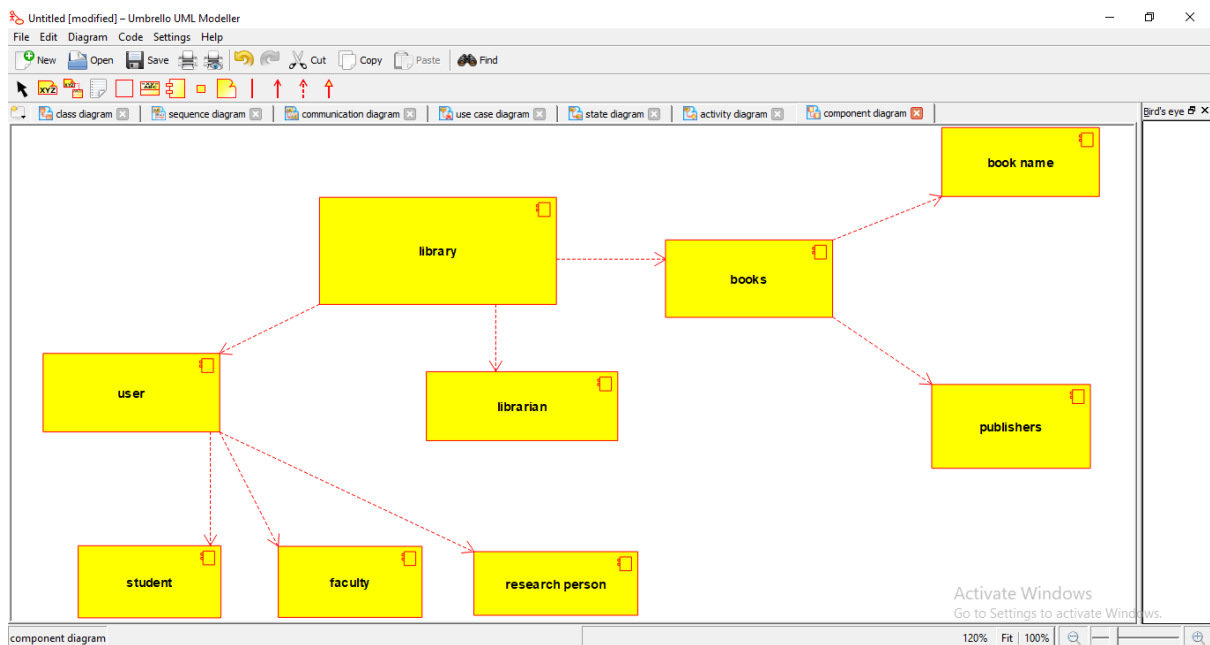
SEQUENCE DIAGRAM



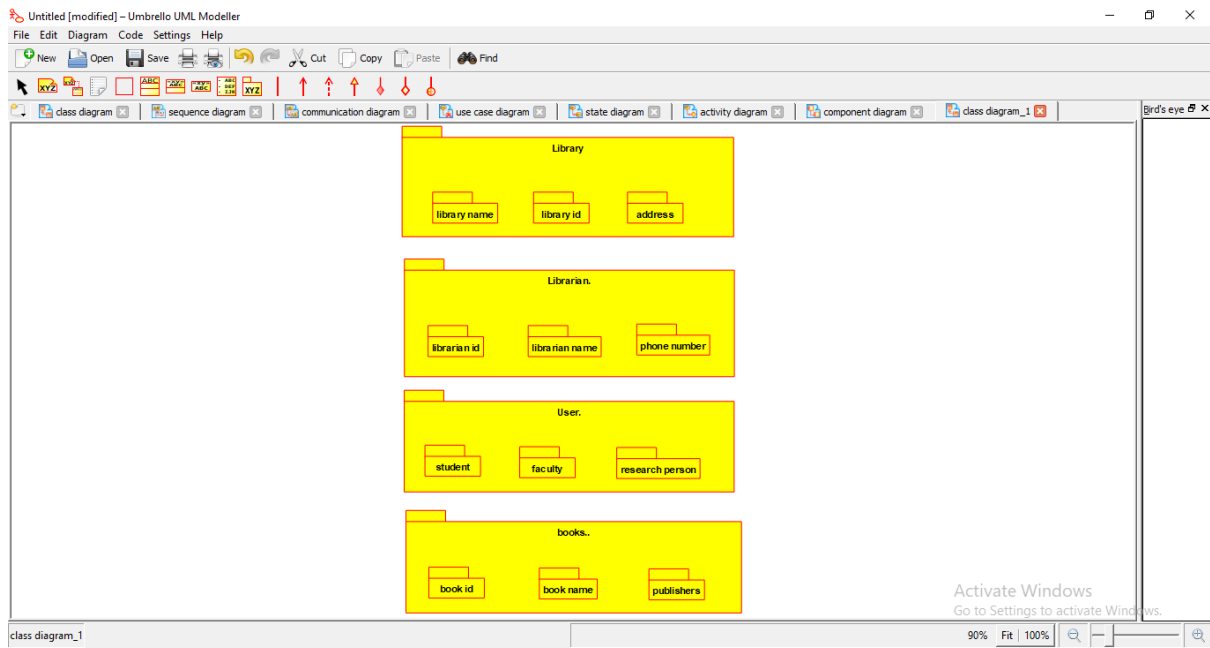
COMMUNICATION DIAGRAM



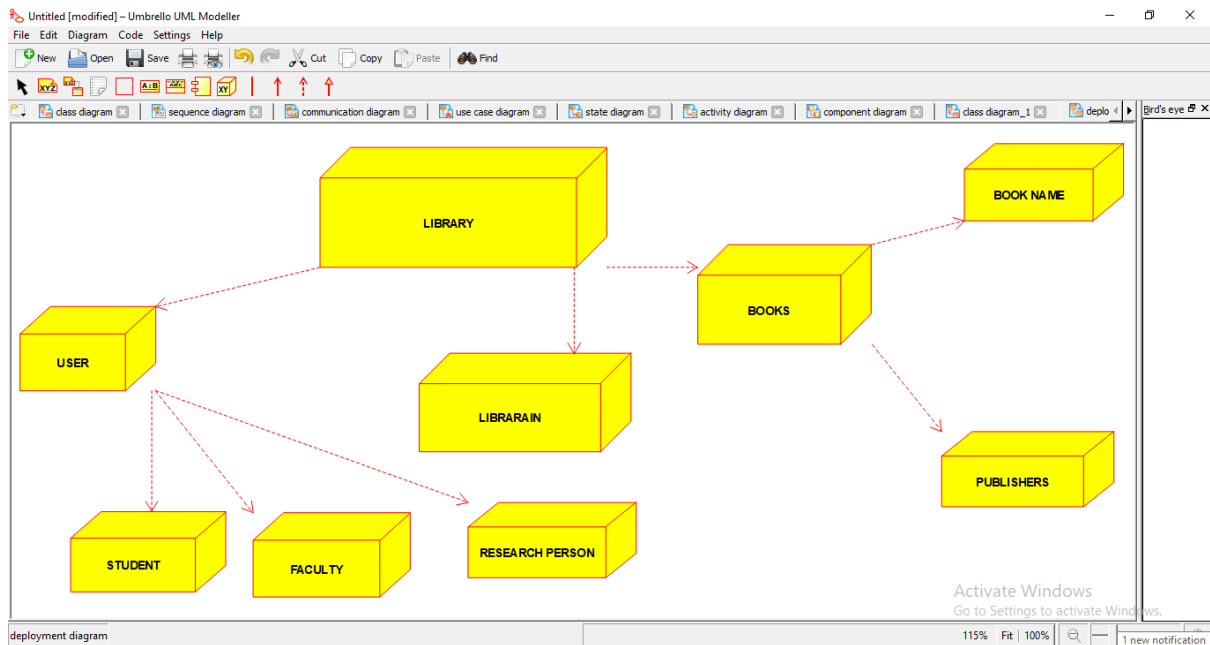
COMPONENT DIAGRAM



PACKAGE DIAGRAM



DEPLOYMENT DIAGRAM



RESULT :- The Use Case Diagram for the Library Management System was successfully designed.